

A01 - Full Validation

Host-side logical NVMe validation framework with deterministic fault injection, queue lifecycle tracking, recovery validation, and firmware-log correlation.

PASS

9	9	9	0
TOTAL	EXECUTED	PASSED	FAILED
0	0	0	2.812 s
ERROR	ABORTED	NOT RUN	DURATION

Suite ID	A01
Suite Name	Full Validation
Generated	2026-09-22 01:10:00
Storage Baseline Reset	true

01

Test Summary

ID	Test	Result	Duration
T01	Device Baseline Check	PASS	0.104 s
T02	Normal Write / Readback	PASS	0.104 s
T03	Invalid Read Range	PASS	0.054 s
T04	Unsupported Command	PASS	0.053 s
T05	Read Timeout + Recovery	PASS	2.078 s
T06	Data Integrity Failure	PASS	0.156 s
T07	Firmware Log Parsing	PASS	0.002 s
T08	Read Error Correlation	PASS	0.052 s
T09	Write Failure + Recovery	PASS	0.208 s

02

Detailed Validation Results

T01Device Baseline CheckPASS

Expected	Device baseline matches configuration
Actual	Baseline validation passed
Validation Checks	10 / 10 PASS
Duration	0.104 s

Key Validation	Expected	Actual	Status
Firmware Revision	1.0.3	1.0.3	PASS
Capacity Blocks	4096	4096	PASS
Logical Block Size	512	512	PASS
Temperature	<= 70	35	PASS
Critical Warning	0	0	PASS
Media Errors	0	0	PASS

T02 Normal Write / Readback**PASS**

Expected	Write and read back T02_WRITE_READBACK at LBA 600
Actual	Write / Readback validation passed
Validation Checks	6 / 6 PASS
Duration	0.104 s

Key Validation	Expected	Actual	Status
Completion Status	SUCCESS	SUCCESS	PASS
Readback LBA	600	600	PASS
Readback Length	1	1	PASS
Readback Data	T02_WRITE_READBACK	T02_WRITE_READBACK	PASS
Unwritten Flag	false	false	PASS

T03 Invalid Read Range**PASS**

Expected	READ LBA 4095, Length 2 returns INVALID_RANGE
Actual	Invalid range correctly rejected
Validation Checks	4 / 4 PASS
Duration	0.054 s

Key Validation	Expected	Actual	Status
Completion Status	INVALID_RANGE	INVALID_RANGE	PASS
Completion Error	INVALID_RANGE	INVALID_RANGE	PASS
Completion Data	None	None	PASS
Timed Out	false	false	PASS

T04 **Unsupported Command****PASS**

Expected	FORMAT returns UNSUPPORTED_COMMAND
Actual	Unsupported command correctly rejected
Validation Checks	5 / 5 PASS
Duration	0.053 s

Key Validation	Expected	Actual	Status
Completion Status	UNSUPPORTED_COMMAND	UNSUPPORTED_COMMAND	PASS
Completion Error	UNSUPPORTED_COMMAND	UNSUPPORTED_COMMAND	PASS
Completion Data	None	None	PASS
Timed Out	false	false	PASS
Submitted Opcode	FORMAT	FORMAT	PASS

T05 **Read Timeout + Recovery****PASS**

Expected	Initial READ times out, Controller Reset clears queue state, and retry succeeds with a new CID
Actual	Timeout / Reset / Retry validation passed
Validation Checks	22 / 22 PASS
Duration	2.078 s

Key Validation	Expected	Actual	Status
Host Timeout Status	TIMEOUT	TIMEOUT	PASS
Outstanding CID After Timeout	[1]	[1]	PASS
SQ State After Timeout	[1, 1]	[1, 1]	PASS
CQ State After Timeout	[0, 0]	[0, 0]	PASS
CID Preserved Across Reset	2	2	PASS
Retry CID	2	2	PASS
Retry Readback Data	TEST_PATTERN_A	TEST_PATTERN_A	PASS
Final Next CID	3	3	PASS

T06 Data Integrity Failure**PASS**

Expected	READ completes successfully but Validator detects returned data mismatch
Actual	Data miscompare correctly detected
Validation Checks	8 / 8 PASS
Duration	0.156 s

Key Validation	Expected	Actual	Status
Fault Read LBA	700	700	PASS
Data Miscompare Detection	!= T06_EXPECTED_DATA	TEST_PATTERN_B	PASS
Stored Data Preserved	T06_EXPECTED_DATA	T06_EXPECTED_DATA	PASS
Stored Data LBA	700	700	PASS

T07 Firmware Log Parsing**PASS**

Expected	Firmware log entries are parsed into structured fields correctly
Actual	Firmware log parsing passed
Validation Checks	10 / 10 PASS
Duration	0.002 s

Key Validation	Expected	Actual	Status
Parsed Entry Count	10	10	PASS
Required Fields Present	true	true	PASS
CID 5 Opcode	READ	READ	PASS
CID 5 LBA	300	300	PASS
CID 5 Error	NAND_READ_FAIL	NAND_READ_FAIL	PASS
LBA 2000 Error	NAND_PROGRAM_FAIL	NAND_PROGRAM_FAIL	PASS

T08 Read Error Correlation**PASS**

Expected	NAND_READ_FAIL completion correlates with firmware log by CID, LBA, status, and error
Actual	Read error correlation passed
Validation Checks	12 / 12 PASS
Duration	0.052 s

Key Validation	Expected	Actual	Status
Completion Status	FAILED	FAILED	PASS
Completion Error	NAND_READ_FAIL	NAND_READ_FAIL	PASS
CID Correlation Count	1	1	PASS
LBA Correlation Count	1	1	PASS
Log CID	1	1	PASS
Log LBA	300	300	PASS
Log Error	NAND_READ_FAIL	NAND_READ_FAIL	PASS

T09 Write Failure + Recovery**PASS**

Expected	NAND_PROGRAM_FAIL preserves old data, correlates with firmware log, and succeeds after reset and retry
Actual	Write failure / recovery validation passed
Validation Checks	28 / 28 PASS
Duration	0.208 s

Key Validation	Expected	Actual	Status
Program Failure Error	NAND_PROGRAM_FAIL	NAND_PROGRAM_FAIL	PASS
Storage Unchanged	OLD_PATTERN	OLD_PATTERN	PASS
CID Preserved Across Reset	3	3	PASS
Retry Write CID	3	3	PASS
Readback CID	4	4	PASS
Recovered Data	T09_RECOVERED_DATA	T09_RECOVERED_DATA	PASS
Final SQ State	[2, 2]	[2, 2]	PASS
Final CQ State	[2, 2]	[2, 2]	PASS
Final Next CID	5	5	PASS

03

Failure and Recovery Evidence

T05 - READ Timeout + Recovery

Initial Command	CID 1 / READ / LBA 100
Host Result	TIMEOUT
Completion	None
Outstanding After Timeout	[1]
Reset Queue	SQ 1/1 -> 0/0, CQ 0/0 -> 0/0
CID After Reset	2
Retry	CID 2 / SUCCESS / TEST_PATTERN_A

T06 - Data Miscompare Detection

Fault Command	CID 2 / READ / LBA 700
CQ Status	SUCCESS
Returned Data	TEST_PATTERN_B
Stored Data Verification	T06_EXPECTED_DATA
Interpretation	I/O completed successfully; validator detected returned-data mismatch while stored data remained intact.

T08 - NAND Read Failure Correlation

Host Command	CID 1 / READ / LBA 300
CQ Completion	FAILED / NAND_READ_FAIL
Firmware Log CID	1
Firmware Log LBA	300
Firmware Log Error	NAND_READ_FAIL
Correlation	CID + Opcode + LBA + Status + Error matched

T09 - Program Failure + Recovery

Failed WRITE	CID 1 / FAILED / NAND_PROGRAM_FAIL
Storage After Failure	OLD_PATTERN
Reset Preserved CID	3
Retry WRITE	CID 3 / SUCCESS
Final READ	CID 4 / T09_RECOVERED_DATA
Final Queue	SQ 2/2, CQ 2/2, next CID 5

Technical Notes and Scope

Queue Model	The project models Submission Queue (SQ), Completion Queue (CQ), CID allocation, queue head/tail progression, and logical doorbell events on the Host side.
Mock Controller	The controller executes in a separate Python subprocess and receives Host-assigned commands and CIDs.
Fault Injection	Timeout, NAND_READ_FAIL, NAND_PROGRAM_FAIL, and data miscompare faults are deterministic and designed for validation workflow demonstration.
Firmware Log	Runtime firmware-style log entries are generated by the Mock Controller process and can be correlated with command results by CID, LBA, status, and error.
STOP v1	Cancellation is cooperative. Active fake_nvme subprocesses can be terminated and recovered through simulated controller reset; arbitrary Python deadlocks or non-cooperative infinite loops are outside the v1 stop guarantee.
Hardware Scope	This project is a Host-side logical NVMe validation simulation. It does not emulate or claim real PCIe MMIO, DMA, PRP/SGL, MSI/MSI-X, FTL, NAND media behavior, or physical SSD firmware validation.

Purpose: demonstrate validation design, automation structure, failure handling, recovery logic, queue-state reasoning, and evidence correlation in a deterministic mock environment.